HCMv3 Developer Setup

How to build and run Web.Host and Angular on a new machine · Gulf Business Solutions · October 2026

1. Install once

Tool	Version	Notes
.NET SDK	**10.0**	Check with `dotnet --list-sdks`
Visual Studio 2026 or Rider	Latest	Optional, but the easiest way to run Web.Host
SQL Server Express	Any recent	The instance must be named `SQLEXPRESS`
DevExpress	**23.2**	**Required.** Install with the company licence. The Reports project uses DevExpress 23.2.13 and the installer adds the local package feed. Without it the build fails with "package DevExpress… not found".
Node.js	**22.x**	
Yarn	1.22	`npm i -g yarn` — the project uses `yarn.lock`
Git / GitHub Desktop		Access to `Gulf-Business-Solutions-GBS/HCMv3`
Ollama	Optional	Only for the AI assistant widget (`localhost:11434` , model `hcmv3-qwen3-4b`)

2. Get the code

```
git clone https://github.com/Gulf-Business-Solutions-GBS/HCMv3.git C:\GBS\HCM\HCMv3
```

3. Database

The app connects to `Server=localhost\SQLEXPRESS; Database=HCMv3` with Windows login (no password).

Option A (recommended): restore the scrambled backup. Get `HCMv3-scrambled.bak` from the team lead (shared privately, never through GitHub). Restore it in SQL Server Management Studio as database **`HCMv3`** . It contains tenants, setup and anonymised employees. Then run the Migrator below so the database matches the latest code.

Option B: empty database. Just run the Migrator. It creates the database and the host admin (`admin` / `123qwe`), but no HR data.

```
dotnet run --project C:\GBS\HCM\HCMv3\aspnet-core\src\HCMv3.Migrator
```

Press **Y** when it asks to migrate.

4. Start Angular first

```
cd C:\GBS\HCM\HCMv3\angular
yarn
yarn create-dynamic-bundles
yarn start
```

Wait until it says it has compiled and is listening on **http://localhost:4200**. The first build takes 5–10 minutes.

5. Then start Web.Host

- **Visual Studio:** open `aspnet-core\HCMv3.All.sln`, set **HCMv3.Web.Host** as the startup project, choose the **HCMv3.Web.Host** profile and press **F5**. It runs on **https://localhost:44301** and opens Swagger.
- **Command line:**

```
dotnet run --project C:\GBS\HCM\HCMv3\aspnet-core\src\HCMv3.Web.Host --launch-profile
HCMv3.Web.Host
```

6. Open the app

Go to **http://localhost:4200** and log in.

Common errors and fixes

Error	Cause	Fix
`DevExpress.*` not found during restore	DevExpress 23.2 not installed	Install DevExpress 23.2
Cannot reach the API / certificate error in browser	Dev HTTPS certificate not trusted	`dotnet dev-certs https --trust`, then open https://localhost:44301 once and accept
Web.Host connects to the wrong or production database	Started with `--no-launch-profile`	Always use the launch profile (or F5), which sets `Development`
Angular: `Property '… ServiceProxy' does not exist`	Proxies out of date after a backend change	Run Web.Host, then in `angular\nswag` run `node ../node_modules/nswag/bin/nswag.js run`, then restart `yarn start`
Port 4200 in use, or Angular hangs	Two `ng serve` running	Run only one Angular dev server at a time
Angular build dies (out of memory)	Less than 16 GB RAM	Start Angular before Web.Host; close other heavy apps
AI assistant widget shows an error	Ollama not running	Ignore it, or install Ollama and pull the model

Working together

- Work on your own branch and open a pull request into `main`; don't push straight to `main`.
- Pull `main` before starting work each day.
- If you change the backend API (new service or DTO), regenerate the Angular proxies and commit `angular/src/shared/service-proxies/service-proxies.ts` in the same commit.
- Never commit passwords, connection strings to real servers, or database backups.